Tutorial Sheet 8: Eigenvalues and eigenspaces

Getting started.

1. (a) We compute the row echelon form of $A - 2I_3$:

$$A - 2I_3 = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 6 & -11 & 4 \end{pmatrix} \rightsquigarrow \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & -8 & 4 \end{pmatrix} \rightsquigarrow \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

There is exactly one free variable, so 2 is an eigenvalue of A and the dimension of the associated eigenspace is 1. We compute a basis of the eigenspace by finding the general solution, so we set $z = \alpha$, and by substitution, we get $y = \alpha/2$ and $x = \alpha/4$. So the general solution is of the form

$$\alpha \begin{pmatrix} 1/4 \\ 1/2 \\ 1 \end{pmatrix}$$

and it follows that a basis of $E_1(A)$ is $(1/4, 1/2, 1)^T$.

(b) We compute the row echelon form of $A - 4I_3$:

$$A - 4I_3 = \begin{pmatrix} -4 & 1 & 0 \\ 0 & -4 & 1 \\ 6 & -11 & 2 \end{pmatrix} \rightsquigarrow \begin{pmatrix} -4 & 1 & 0 \\ 0 & -4 & 1 \\ 0 & -19 & 4 \end{pmatrix} \rightsquigarrow \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

There is no free variable, so 4 is not an eigenvalue of A.

2. We consider the following matrix:

$$A = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}.$$

(a) We have

$$A - I_4 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

We could compute the row echelon form and find the number of free variables as usual (and you should do it, that is good practise!), but we can avoid computations altogether here, here is how. The first two columns are linearly independent, so

$rk(A - I_4) \geq 2$. Moreover, columns 3 and 4 are identical to columns 2 and 1 respectively, so contribute nothing to the rank, so $\mathrm{rk}(A - I_4) \leq 2$. Thus the rank of $A - I_4$ is exactly 2, and by the Rank-Nullity Theorem, the dimension of $N(A - I_4) = E_1(A)$ is $4 - 2 = 2$.

(b) We have

$$A - 3I_4 = \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

We could compute the row echelon form and find the number of free variables as usual (and again, do it!), but we can avoid computations altogether here again. The first two columns are linearly independent, so $rk(A - 3I_4) \geq 2$. Moreover, columns 3 and 4 are multiples of columns 2 and 1 respectively, so contribute nothing to the rank, so $\mathrm{rk}(A - 3I_4) \leq 2$. Thus the rank of $A - 3I_4$ is exactly 2, and by the Rank-Nullity Theorem, the dimension of $N(A - 3I_4) = E_3(A)$ is $4 - 2 = 2$.

(c) We have $\dim E_1(A) + \dim E_3(A) = 4$ and A is a 4×4 matrix, so by a result of the course A is diagonalisable, and its eigenvalues are 1 and 3.

Problems.

3. **A linearly independent family.**

(a) Let $u = (u_n), v = (v_n)$ be two real requences, and let $\lambda \in \mathbb{R}$. We have

$$S(u+v) = S\begin{pmatrix} u_0 + v_0 \\ u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \\ u_4 + v_4 \\ \vdots \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ \vdots \end{pmatrix} = S(u) + S(v).$$

Moreover,

$$S(\lambda u) = S\begin{pmatrix} \lambda u_0 \\ \lambda u_1 \\ \lambda u_2 \\ \lambda u_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} \lambda u_1 \\ \lambda u_2 \\ \lambda u_3 \\ \lambda u_4 \\ \vdots \end{pmatrix} = \lambda \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \end{pmatrix} = \lambda S(u).$$

Thus, S is a linear map.

(b) For $\lambda > 0$, we have

$$S(u^\lambda) = S\begin{pmatrix} 1 \\ \lambda \\ \lambda^2 \\ \lambda^3 \\ \vdots \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda^2 \\ \lambda^3 \\ \lambda^4 \\ \vdots \end{pmatrix} = \begin{pmatrix} \lambda \times 1 \\ \lambda \times \lambda \\ \lambda \times \lambda^2 \\ \lambda \times \lambda^3 \\ \vdots \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ \lambda \\ \lambda^2 \\ \lambda^3 \\ \vdots \end{pmatrix} = \lambda u^\lambda.$$

Since u^λ is not the zero-vector, it follows that u^λ is an eigenvector of S for the eigenvalue λ.

(c) By (b), the family $u^{\lambda_1}, \ldots, u^{\lambda_k}$ is a family of eigenvectors of S for pairwise distinct eigenvalues. By a theorem from the course, this implies that this family is linearly independent.

4. **A diagonalisation problem.**

(a) All the columns are multiples of the first column (and are non-zero), so the rank is 1. Alternatively, you can compute the row echelon form and check that there is only one pivot variable to deduce that the rank is 1.

(b) By the Rank-Nullity Theorem, we have $\dim N(A) = n - \operatorname{rk}(A) = n - 1$. But $N(A) = N(A - 0I_n) = E_0(A)$, so $\dim E_0(A) = n - 1 > 0$, so 0 is indeed an eigenvalue, and the dimension of the associated eigenspace is $n - 1$.

(c) We have

$$A\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} n \\ \vdots \\ n \end{pmatrix} = n\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix},$$

so $(1, \ldots, 1)^T$ is an eigenvector of A for the eigenvalue n, and so $\dim E_n(A) \geq 1$.

(d) We have $\dim E_0(A) + \dim E_n(A) \geq (n-1)+1 = n$. So by a theorem of the course, it follows that A is diagonalisable with eigenvalue 0 (and associated eigenspace of dimension $n - 1$) and n (and associated eigenspace of dimension 1).

5. **Unique eigenvalues.**

(a) Since A is diagonalisable with eigenvalue λ only, there exists an invertible $n \times n$ matrix P such that we have

$$A = P\begin{pmatrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{pmatrix} P^{-1} = \lambda P I_n P^{-1} = \lambda P P^{-1} = \lambda I_n.$$

(b) We have

$$A - \lambda I_3 = \begin{pmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{pmatrix},$$

which is already in row echelon form and has a free variable if and only if $\lambda = 1$. (If you have seen the section on characteristic polynomial already, there is a shorter proof: the matrix is upper triangular, so its eigenvalues are exactly its diagonal coefficients, so only 1.)

As this is its only eigenvalue and $A \neq I_3$, it follows from (a) that A is not diagonalisable.

6. **Nilpotent matrices.**

(a) Let $\vec{x} \neq$ be an eigenvector of A for the eigenvalue λ. We show by induction on $k \geq 1$ that $\vec{x}$ is an eigenvector of A^k for the eigenvalue λ^k. . This is true for $k = 1$ by assumption, so let us assume that the result is true for k, and let us prove it for $k + 1$. We have

$$A^{k+1}\vec{x} = A(A^k\vec{x}) = A(\lambda^k\vec{x}) = \lambda^k A\vec{x} = \lambda^k\lambda\vec{x} = \lambda^{k+1}\vec{x},$$

which concludes the induction.

(b) We check that:

$$A^2 = \begin{pmatrix} 12 & -4 & -4 \\ 12 & -4 & -4 \\ 24 & -8 & -8 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

so A is nilpotent.

(c) Let λ be an eigenvalue of a nilpotent matrix A. Then there exists $k \geq 1$ such that $A^k = 0_n$. We know from (a) that λ^k is an eigenvalue of $A^k = 0_n$. As the only eigenvalue of 0_n is 0, we have $\lambda^k = 0$, hence $\lambda = 0$.

Let us now show that 0 is indeed an eigenvalue of A. We have $A^k = 0_n$, hence $\det(A^k) = 0 = \det(A)^k$, hence $\det(A) = 0$. So A is not invertible and it follows that there exists a vector $\vec{x} \neq \vec{0}$ such that $A\vec{x} = \vec{0} = 0 \times \vec{x}$. Thus, 0 is an eigenavlue of A (and a corresponding eigenvector is $\vec{x}$.)

(d) Suppose that A is nilpotent and diagonalisable. Since 0 is the only possible eigenvalue of A by (b), there exists an invertible $n \times n$ matrix P such that

$$A = P \begin{pmatrix} 0 & & \\ & \ddots & \\ & & 0 \end{pmatrix} P^{-1} = P 0_n P^{-1} = 0_n.$$